

Java Quiz



Java Step 1 Quiz

Intro to Java · The Basics · Looping & Branching



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Question 1

Java

What is Java best described as?

- 1 Only a programming language, nothing more
- 2 A programming language, a platform (the JRE), and a library of classes
- 3 A database management system
- 4 A web browser plugin



Question 1

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✓ **Answer 2 — Java is three things at once — a language you write in, a platform (the JRE) code runs on, and a large built-in library of classes grouped into packages**



Why can the same compiled Java program run unmodified on Windows, macOS, and Linux?

- 1 Because Java source code is never compiled
- 2 Because every operating system secretly runs Windows underneath
- 3 Because javac compiles source into platform-independent bytecode, and any platform's JVM can run it
- 4 Because Java programs are actually written directly in machine code



Question 2

Java

Why can the same compiled Java program run unmodified on Windows, macOS, and Linux?

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2 Because every operating system secretly runs Windows underneath

3 Because javac compiles source into platform-independent bytecode, and any platform's JVM can run it ✓

4 Because Java programs are actually written directly in machine code

✓ Answer 3 — javac produces .class bytecode that isn't tied to any OS; each platform just needs its own JVM to run that same bytecode — write once, run anywhere



What are the three steps to get from a `.java` source file to a running program?

1 Write the `.java` file, run it directly, then compile it afterwards for speed

2 Write the `.java` file, compile it with `javac` into a `.class` file, then run the `.class` file inside the JRE with `java`

3 Write the `.java` file, upload it to Oracle, and download a compiled binary

4 Write the `.java` file and email it to the JVM



Question 3

Java

What are the three steps to get from a `.java` source file to a running program?

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2 Write the `.java` file, compile it with `javac` into a `.class` file, then run the `.class` file inside the JRE with `java` ✓

3 Write the `.java` file, upload it to Oracle, and download a compiled binary

4 Write the `.java` file and email it to the JVM

✓ Answer 2 — `javac MyCode.java` produces `MyCode.class`; `java MyCode` then runs that bytecode inside the JRE



Question 4

Java

Besides primitives, what are the other two categories of data type in Java?

1 Constants and variables

2 Structs and enums

3 Object references and arrays

4 Integers and decimals



Question 4

Java

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1 Constants and variables

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4 Integers and decimals

✓ Answer 3 — Java groups its data types into three categories — primitives, object references, and arrays



Question 5

Java

What is the default type of a whole-number literal like `12` ?

1 long

2 int

3 short

4 byte



Question 5

Java

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1 long

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3 short

4 byte

✓ Answer 2 — Whole number literals default to int — you need an L suffix (`12L`) to force a long



Question 6

Java

Given `byte b = 12, c = 10;` , **why does** `byte d = b + c;` **fail to compile?**

1 byte variables can't be added together

2 Java doesn't support the + operator on numbers

3 Integer arithmetic always produces an int result, and an int can't be assigned into a byte without an explicit cast

4 b and c must be declared as int first



Question 6

Java

Given `byte b = 12, c = 10;` , **why does** `byte d = b + c;` **fail to compile?**

1 byte variables can't be added together

2 Java doesn't support the + operator on numbers

3 Integer arithmetic always produces an int result, and an int can't be assigned into a byte without an explicit cast 

4 b and c must be declared as int first

 **Answer 3** — The result of `b + c` is always computed as an int; assigning it into a smaller byte variable needs an explicit cast — `byte d = (byte)(b + c);`



Question 7

Java

What does the modulus operator `%` return?

1 The result of a division, rounded up

2 The remainder after a division

3 The average of two numbers

4 A random number



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1 The result of a division, rounded up

2 The remainder after a division



3 The average of two numbers

4 A random number

✓ Answer 2 — `%` gives you the remainder left over after dividing — `5 % 2` is 1



Question 8

Java

In a `switch` statement, what happens if a matching `case` has no `break`?

1 A compile error is thrown

2 The switch exits immediately without running anything

3 Execution falls through and keeps running the statements in the next case(s) until a break is hit

4 Only the default case runs instead



Question 8

Java

In a `switch` statement, what happens if a matching `case` has no `break`?

1 A compile error is thrown

2 The switch exits immediately without running anything

3 Execution falls through and keeps running the statements in the next case(s) until a `break` is hit ☒

4 Only the default case runs instead

☒ Answer 3 — Without `break`, execution falls through into subsequent case blocks regardless of whether their value matches — a common source of bugs



What's the difference between a `while` loop and a `do-while` loop?

1 They are exactly the same

2 `while` checks the condition after the body runs;
`do-while` checks it before

3 `do-while` checks the condition after the body runs, so the body always executes at least once; `while` checks it first and might never run the body

4 `do-while` cannot be used with `break` or `continue`



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4 `do-while` cannot be used with `break` or `continue`

✓ Answer 3 — `do-while` guarantees at least one execution because the check happens at the end of the loop; `while` can skip the body entirely if the condition starts false



What's the difference between `i++` and `++i` ?

- 1 There is no difference — they always behave identically
- 2 `i++` (postfix) uses the current value of `i` and then increments it; `++i` (prefix) increments `i` first and then uses the new value
- 3 `++i` can only be used inside a for loop
- 4 `i++` increments by 2 while `++i` increments by 1



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3 `++i` can only be used inside a for loop

4 `i++` increments by 2 while `++i` increments by 1

✓ Answer 2 — Postfix `i++` returns the old value before incrementing; prefix `++i` increments first and returns the new value — printing them in a loop produces different sequences





Well Done!

That's the end of the Java Step 1 quiz.

What is Java, bytecode & the JVM, compiling & running, data type categories, literals, casting, modulus, switch fall-through, while vs do-while, prefix/postfix: ✓